

EMG Controlled 3D Printed Prosthetic Arm

3D printed arm that is ergonomic, affordable, and fashionable

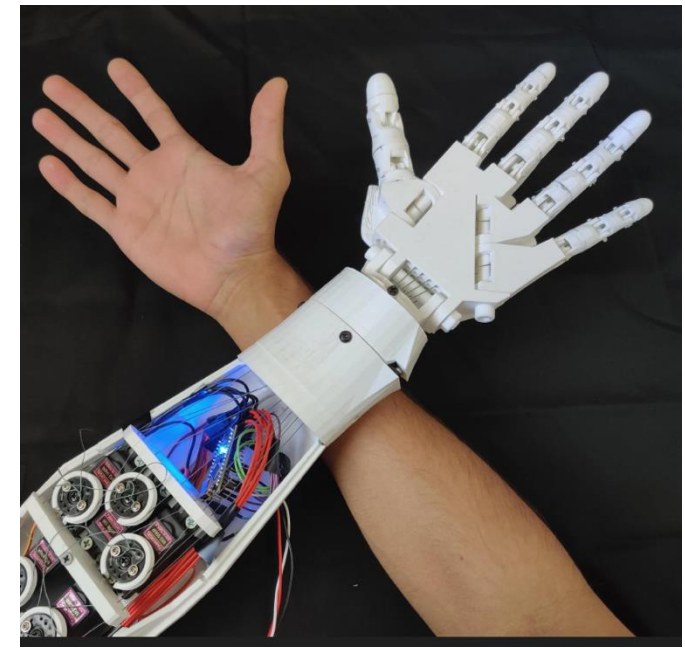
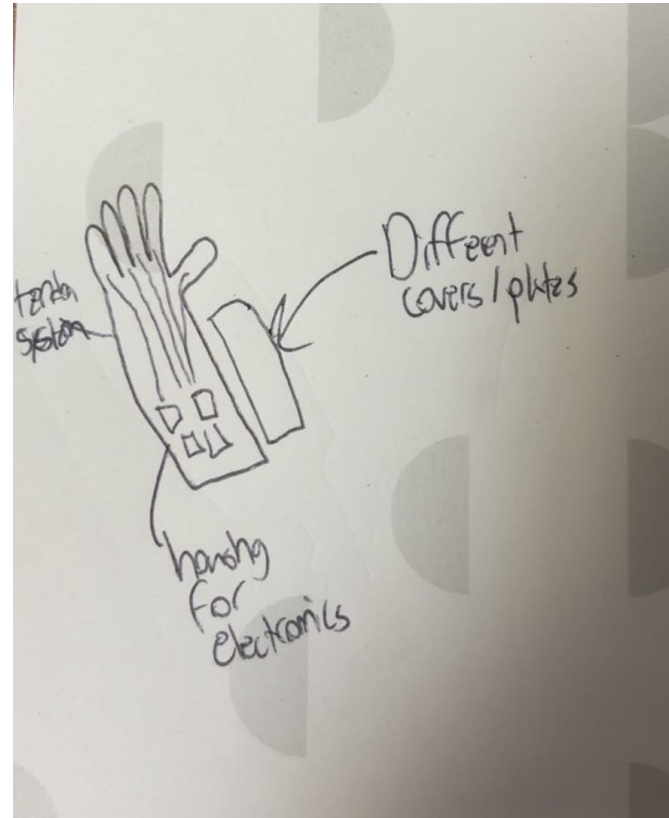
Bryton Montgomery

Problem Context/Intended User

- 1. Real World Problem
 - Very limited prosthetics for people of lower income
 - Many prosthetic limbs are just not fashionable or customizable
 - Modern Prosthetics use mechanical engineering and robotics without customization
- 2. Intended Users
 - low-income individual (13-35)
 - Missing a limb
 - People needing a fashionable and affordable prosthetic/Young Adults
 - Designers and engineers

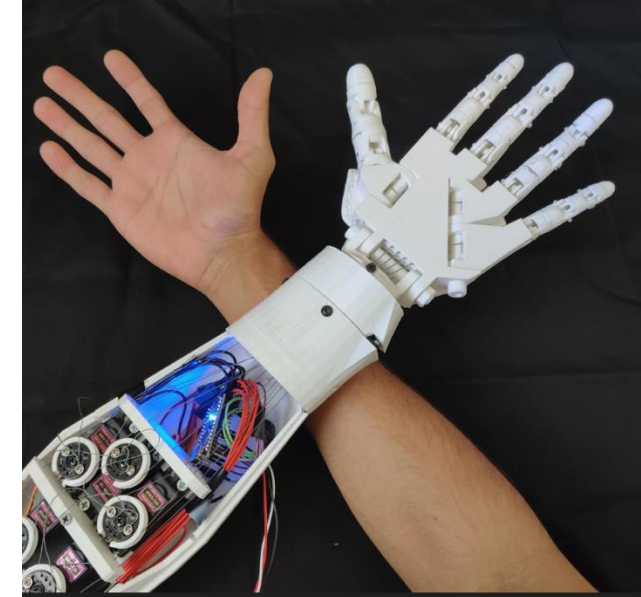
Initial Design Goals

- Make a working EMG controller 3D hand
- Make fashionable 3D printed arm
- Have multiple designs that are swappable at the user's discretion



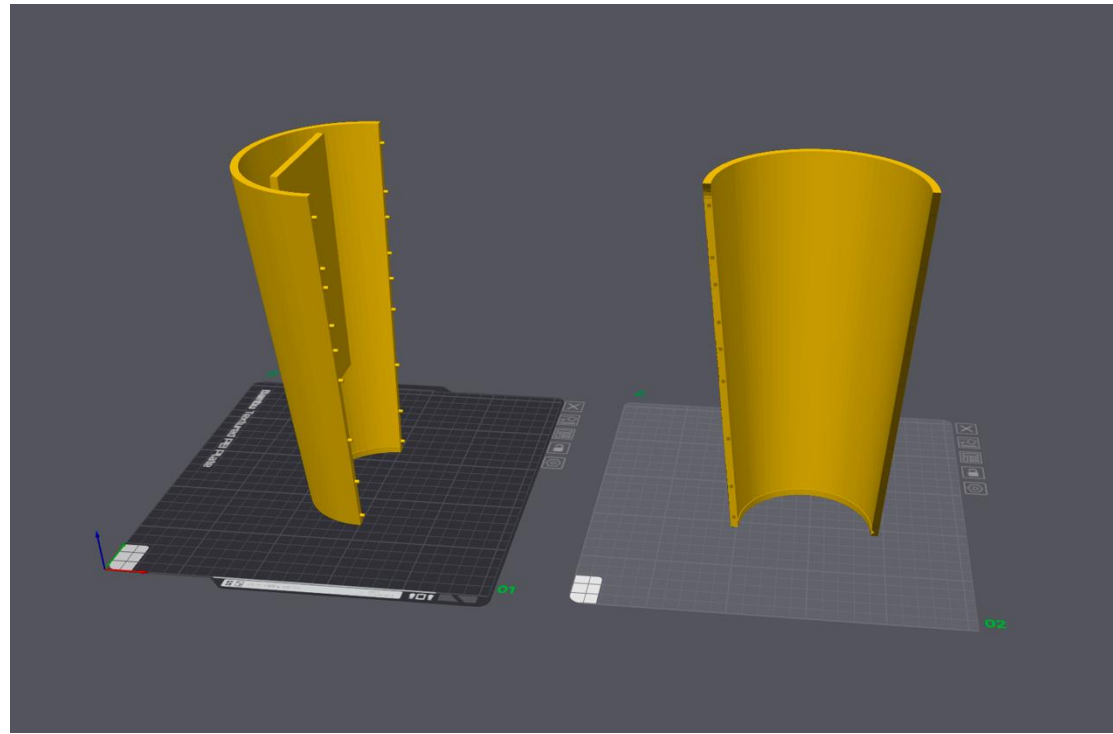
Research/Discovery

- Current prosthetics are too bulky, mechanical looking, and all around not fashionable
- Key Features
 - Articulated fingers
 - Tendon like movement for fingers
 - Adjustable
 - Lightweight yet durable model
- Constraints: size of arm, complexity of design, printing in multiple parts



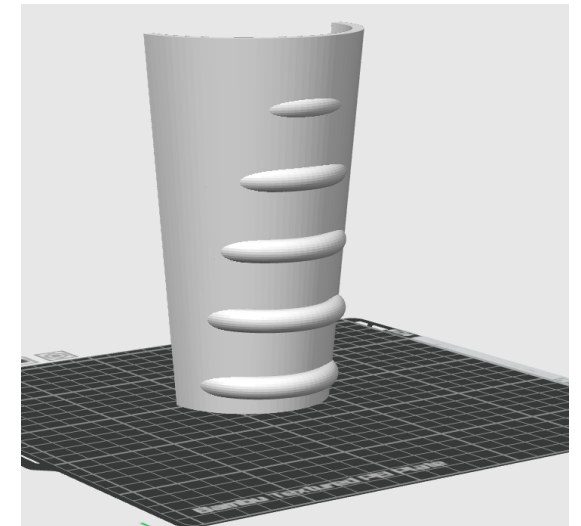
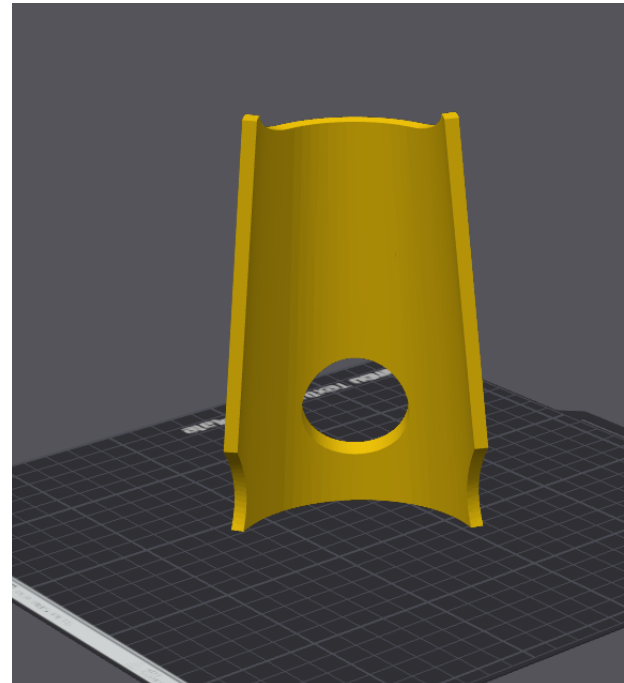
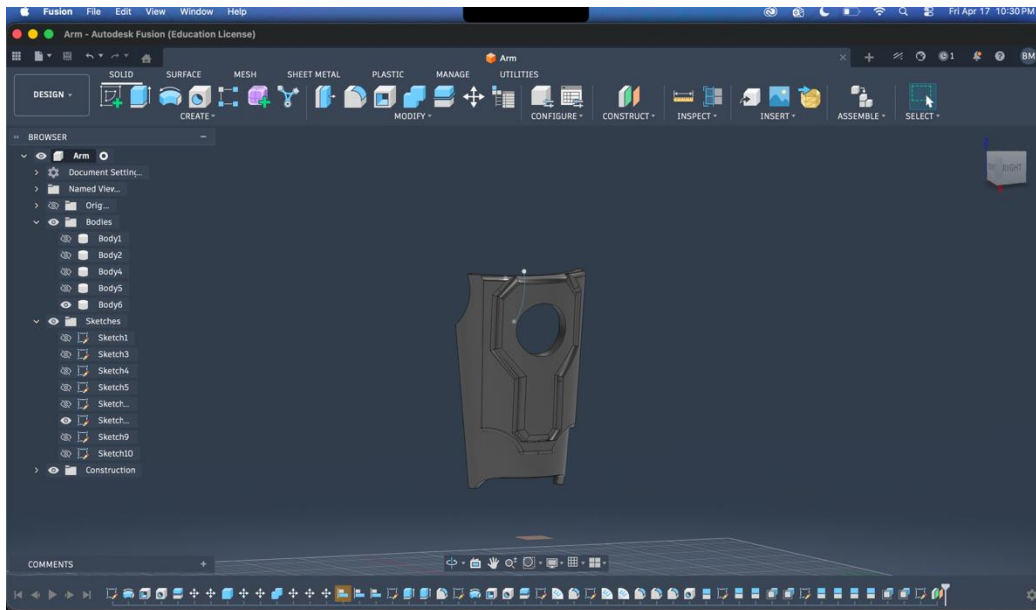
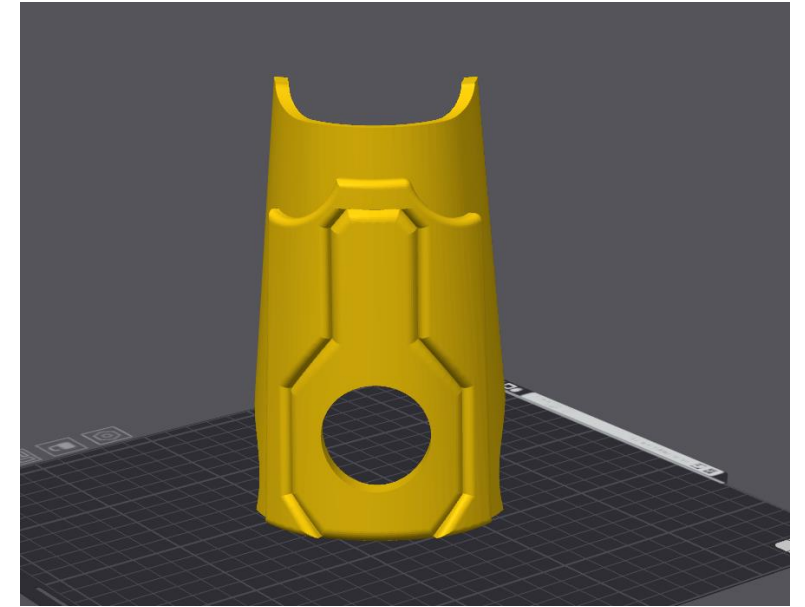
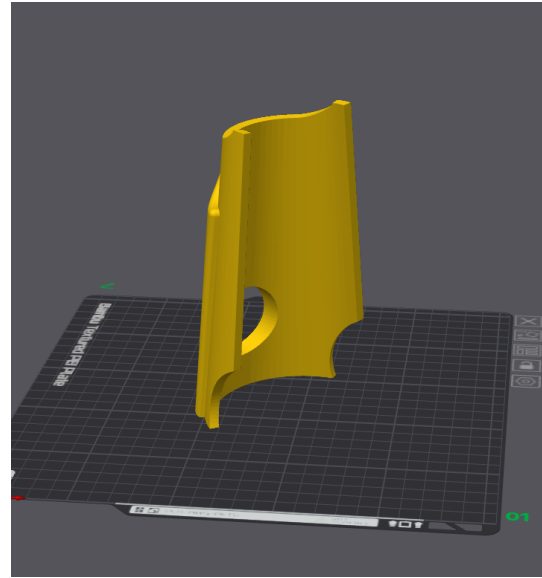
Prototype 1

- Very rough first design
- Too cylindrical
- No design to it
- Margins/Scale was off/Too large
- Description
 - I created a simple model forearm model that detaches and leaves room for electronics



Prototype 2

- Added more design inspiration from Iron Man
 - Added design for light
 - All for electronics to fit
- Proper scaling/Perfect size



Materials and Print Strategy

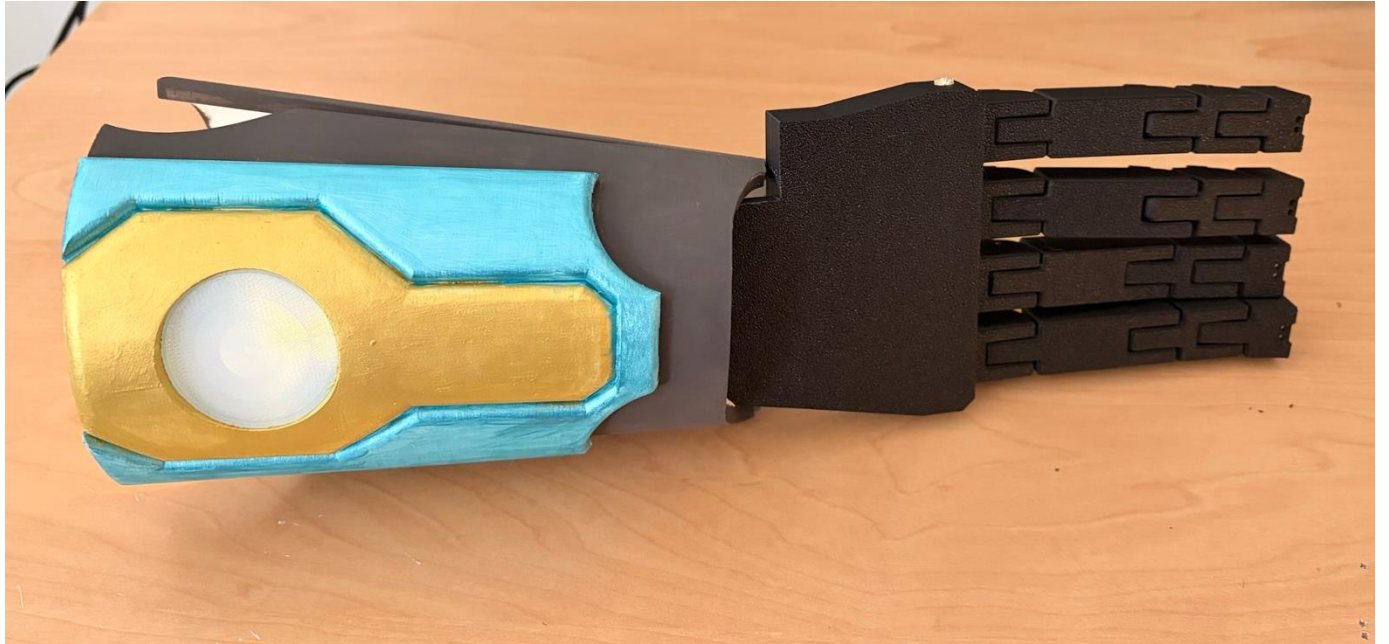
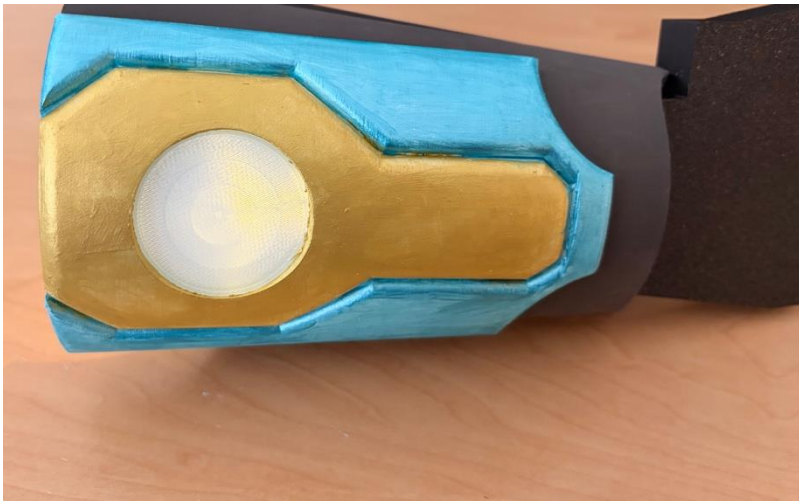
- Printing Strategy
 - Printed upright to reduce the use of supports/support lines
 - Printed in two separate prints to reduce errors
 - Printed in slower speed to ensure quality print
- Materials
 - Standard white PLA for both arm halves
 - Translucent blue PLA for light

Issues

- Fusion360
 - Really hard to do simple tasks
 - Delete and separate objects
- Paint did not adhere well to 3D print
 - Sanded and Primer
- No major printing issues:
 - Improperly spun spool

Post Processing and Final Product





Reflection

- **What worked well?**

- Printing process went smoothly
- Colors work well together
- Size is roughly correct

- **What limitations exist?**

- Lack of software
- Lack of knowledge on modeling/tools
- Printing plate size

- **What would you improve with more time?**

- I would improve my modeling skills
- 3D scan my arm to get proper fit
- Choose different materials so that paint would adhere